


Name: - Niriksha Tiwari

Roll no.: - 2501730353

Course: - B.tech CSE (AI & ML)

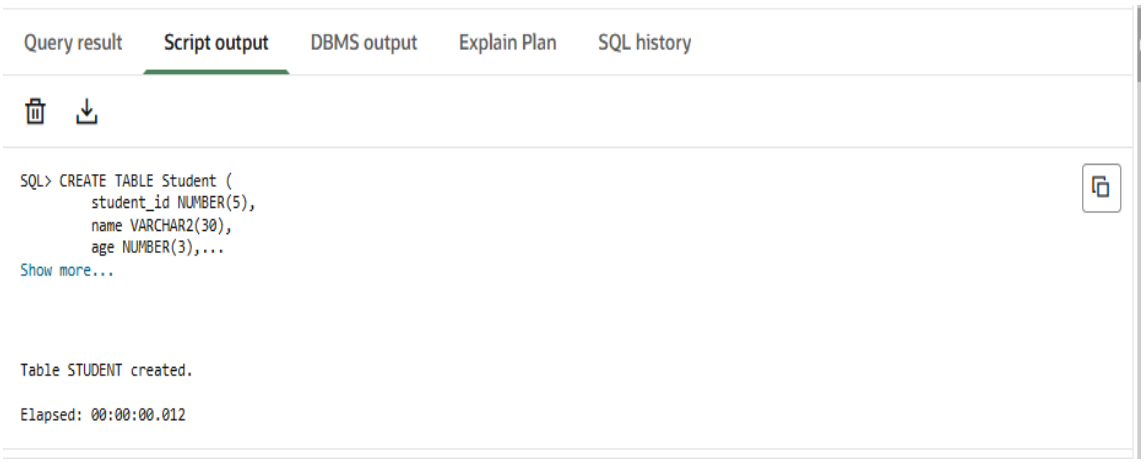
Mentor: - Megha Sharma

Subject: - Database Management System

# ♦ DDL Commands: -

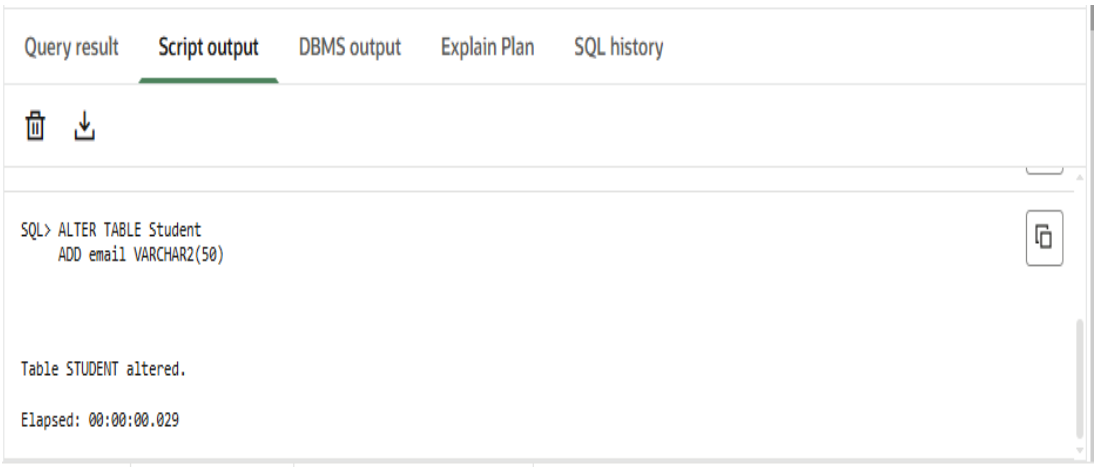
- Create a table: -

```
CREATE TABLE Student (
   student_id NUMBER(5),
   name VARCHAR2(30),
   age NUMBER(3),
   course VARCHAR2(20),
   marks NUMBER(3)
);
```



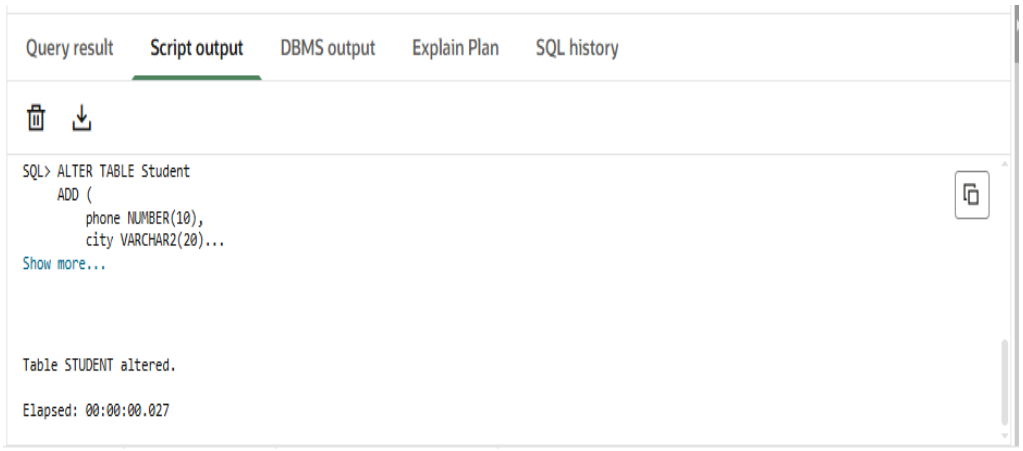

- Alter: -

```
ALTER TABLE Student
ADD email VARCHAR2(50);
```



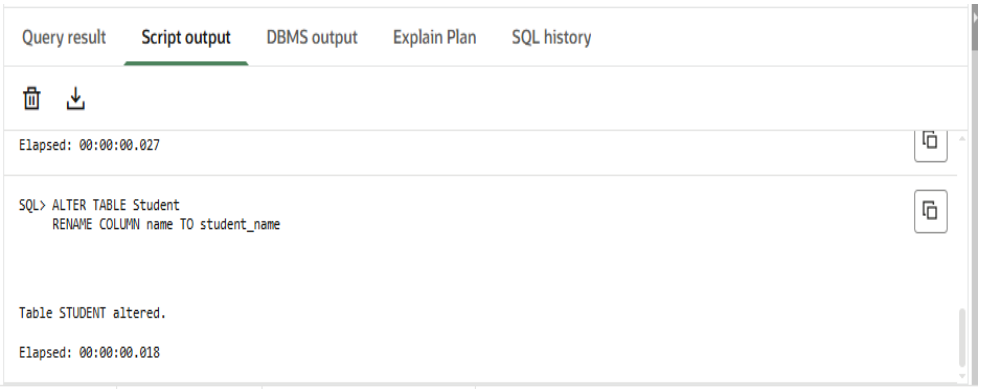

- Adding Multiple Columns: -

```
ALTER TABLE Student
ADD (
  phone NUMBER (10),
  city VARCHAR2(20)
);
```



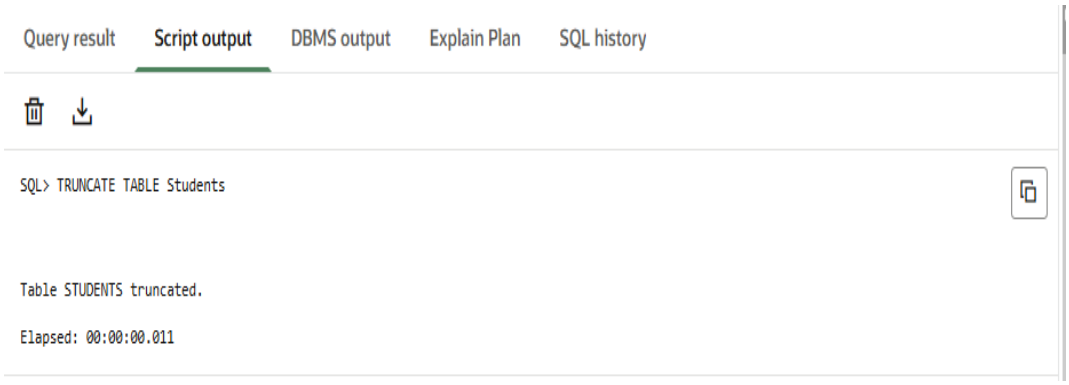

- RENAME a column: -

ALTER TABLE Student
RENAME COLUMN name TO student_name;




- TRUNCATE table: -

TRUNCATE TABLE Student;




- Drop Student table: -

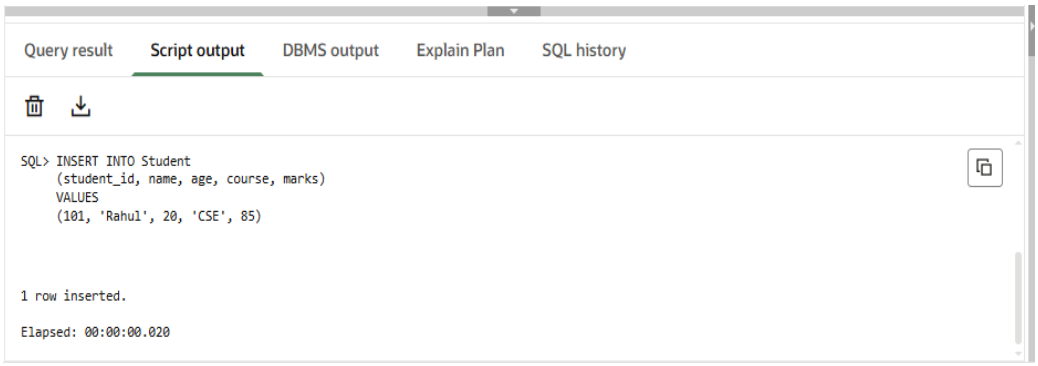
DROP TABLE Students;




# ♦ DML Commands: -

- INSERT: -

```
INSERT INTO Student
(student_id, name, age, course, marks)
VALUES
(101, 'Rahul', 20, 'CSE', 85);
```




- 10 Values in Table: -

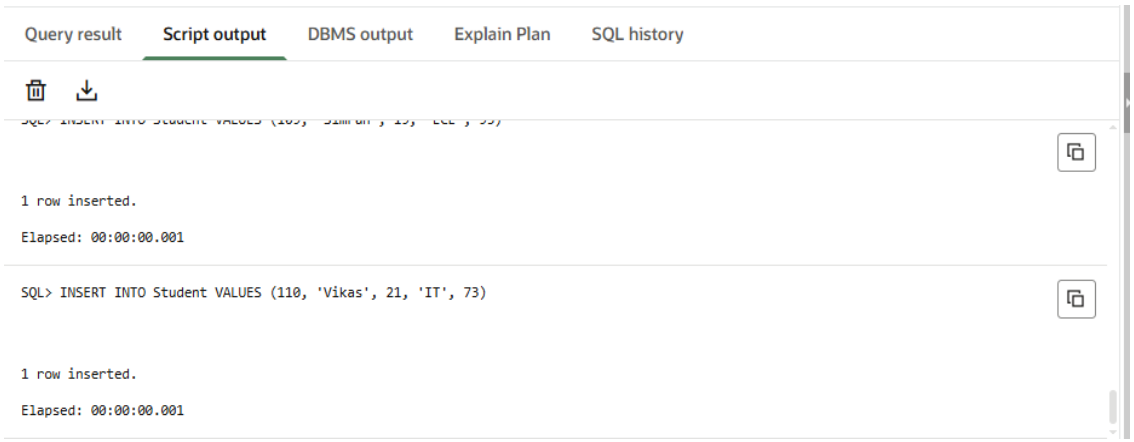
```
INSERT INTO Student
VALUES (101, 'Rahul', 20, 'CSE', 85);

INSERT INTO Student
VALUES (102, 'Priya', 19, 'CSE', 90);

INSERT INTO Student
VALUES (103, 'Aman', 21, 'ECE', 78);

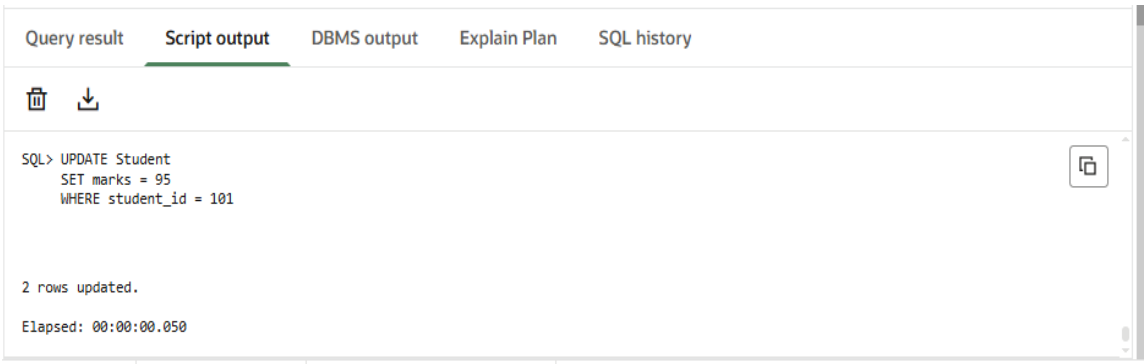
INSERT INTO Student
VALUES (104, 'Neha', 20, 'IT', 88);

INSERT INTO Student
VALUES (105, 'Riya', 19, 'CSE', 92);
  INTO Student
 VALUES (103, 'Aman', 21, 'ECE', 78)
  INTO Student
VALUES (104, 'Neha', 20, 'CSE', 88)
  INTO Student
VALUES (105, 'Riya', 19, 'IT', 92)
SELECT * FROM dual;
```



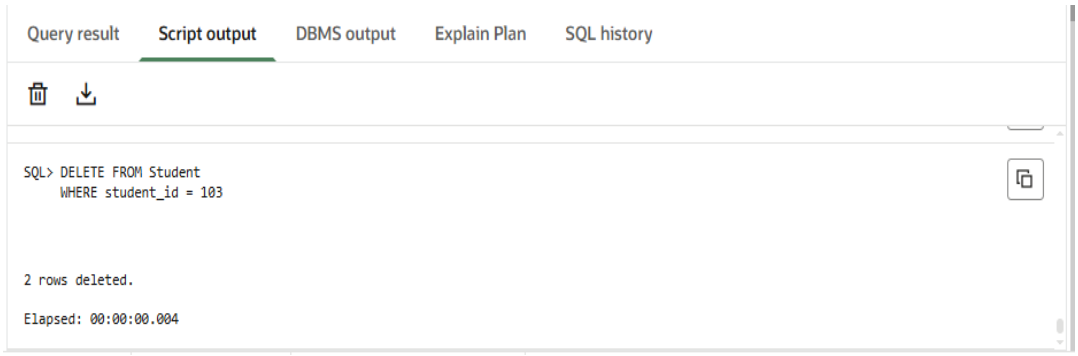

- Update: -

```
UPDATE Student
SET marks = 95
WHERE student_id = 101;
```




- Delete: -

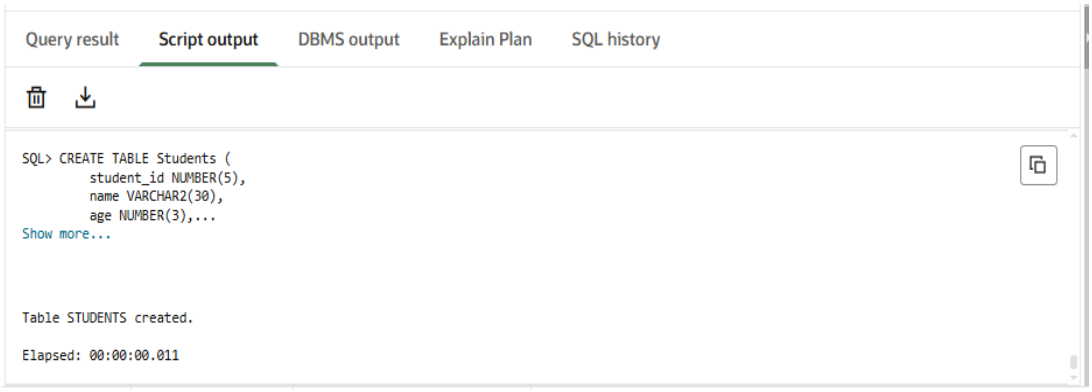
```
DELETE FROM Student
WHERE student_id = 105;
```




- # DQL Commands: -

- CREATE TABLE — DDL: -

```
CREATE TABLE Students (
  student_id NUMBER (5),
  name VARCHAR2(30),
  age NUMBER (5),
  course VARCHAR2(30),
  marks NUMBER (3)
);
```

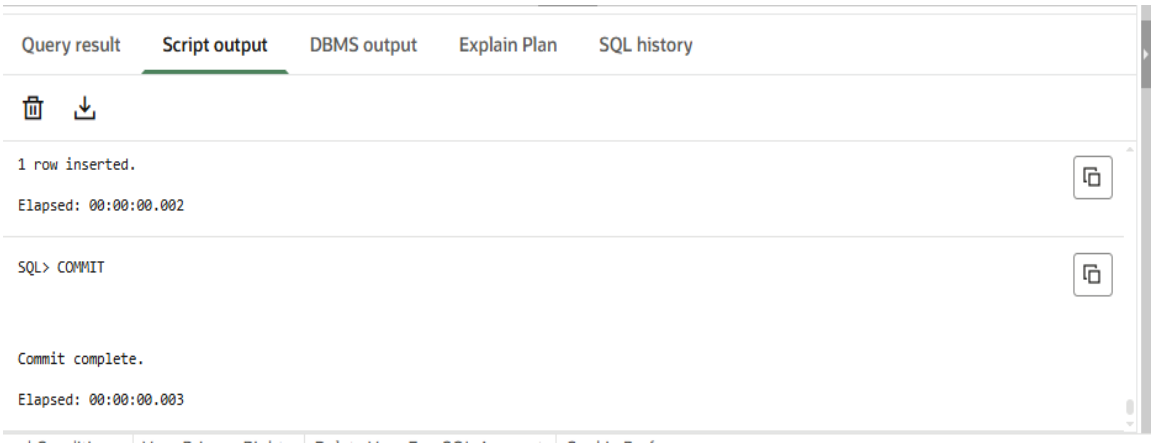



- INSERT 10 ROWS — DDL: -

```
INSERT INTO Students VALUES (101, 'Rahul', 20, 'CSE', 85);
INSERT INTO Students VALUES (102, 'Priya', 19, 'CSE', 90);
INSERT INTO Students VALUES (103, 'Aman', 21, 'ECE', 78);
```

```
INSERT INTO Students VALUES (104, 'Neha', 20, 'IT', 88);
INSERT INTO Students VALUES (105, 'Riya', 19, 'CSE', 92);
INSERT INTO Students VALUES (106, 'Karan', 21, 'ECE', 75);
INSERT INTO Students VALUES (107, 'Anjali', 20, 'IT', 81);
INSERT INTO Students VALUES (108, 'Arjun', 22, 'CSE', 69);
INSERT INTO Students VALUES (109, 'Simran', 19, 'ECE', 95);
INSERT INTO Students VALUES (110, 'Vikas', 21, 'IT', 73);

COMMIT;
```




- Display all records (SELECT): -

```
SELECT * FROM Students;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.002 seconds

| | STUDENT_ID | NAME | AGE | COURSE | MARKS |
|---|---|---|---|---|---|
| 1 | 101 | Rahul | 20 | CSE | 95 |
| 2 | 101 | Rahul | 20 | CSE | 85 |
| 3 | 102 | Priya | 19 | CSE | 90 |
| 4 | 103 | Aman | 21 | ECE | 78 |
| 5 | 104 | Neha | 20 | IT | 88 |
| 6 | 105 | Riya | 19 | CSE | 92 |
| 7 | 106 | Karan | 21 | ECE | 75 |

- Display only student names (SELECT): -

SELECT name FROM Student;

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.008 seconds

| | NAME |
|---|---|
| 1 | Rahul |
| 2 | Rahul |
| 3 | Priya |
| 4 | Aman |
| 5 | Neha |
| 6 | Riya |
| 7 | Karan |

- Display students NOT from CSE (SELECT): -

SELECT * FROM Student
WHERE course != 'CSE';

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.005 seconds

| | STUDENT_ID | NAME | AGE | COURSE | MARKS |
|---|---|---|---|---|---|
| 1 | 103 | Aman | 21 | ECE | 78 |
| 2 | 104 | Neha | 20 | IT | 88 |
| 3 | 106 | Karan | 21 | ECE | 75 |
| 4 | 107 | Anjali | 20 | IT | 81 |
| 5 | 109 | Simran | 19 | ECE | 95 |
| 6 | 110 | Vikas | 21 | IT | 73 |
| 7 | 105 | Riya | 19 | IT | 92 |

- **Group by: -**

[ SQL Worksheet ]*

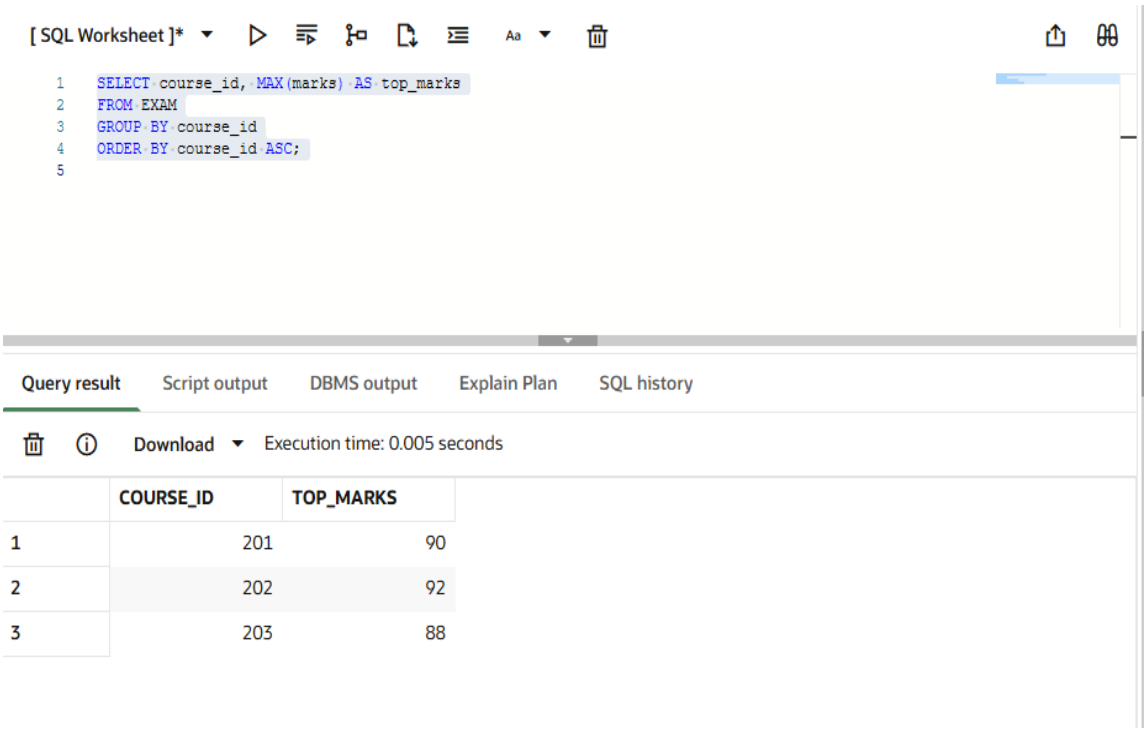
```
1  SELECT course_id, ROUND(AVG(marks), 2) AS average_marks
2  FROM EXAM
3  GROUP BY course_id;SELECT course_id, ROUND(AVG(marks), 2) AS average_marks
4  FROM EXAM
5  GROUP BY course_id;
6
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.001 seconds

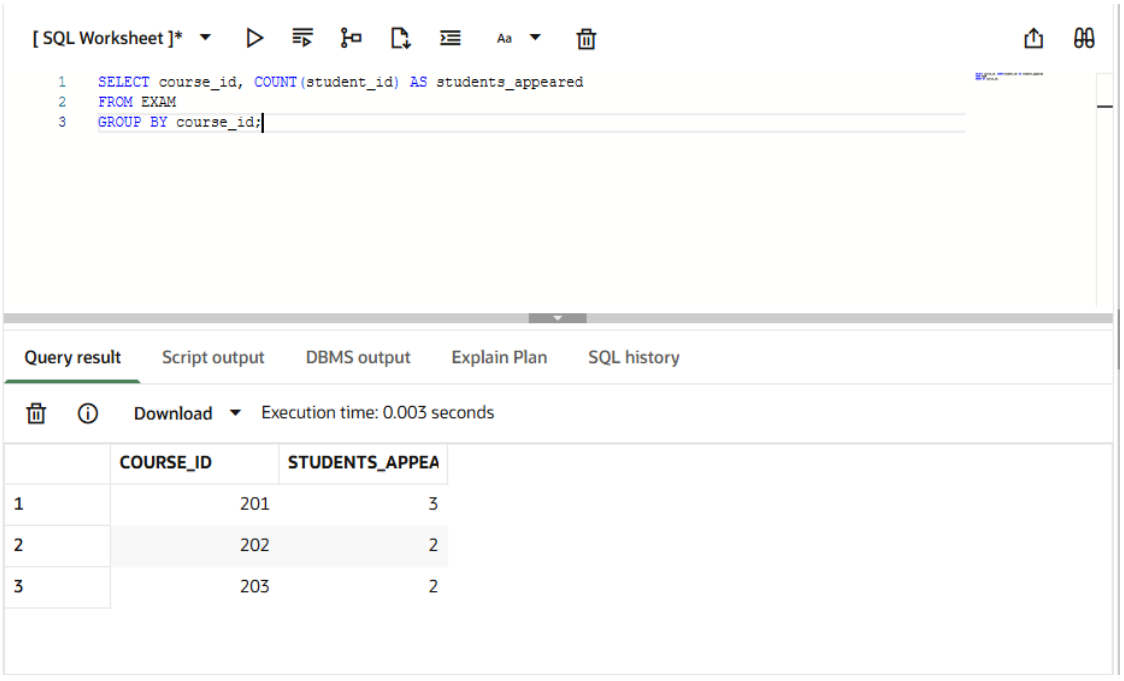
| | COURSE_ID | AVERAGE_MARKS |
|---|---|---|
| 1 | 201 | 84.33 |
| 2 | 202 | 83.5 |
| 3 | 203 | 84 |

- **Order by: -**



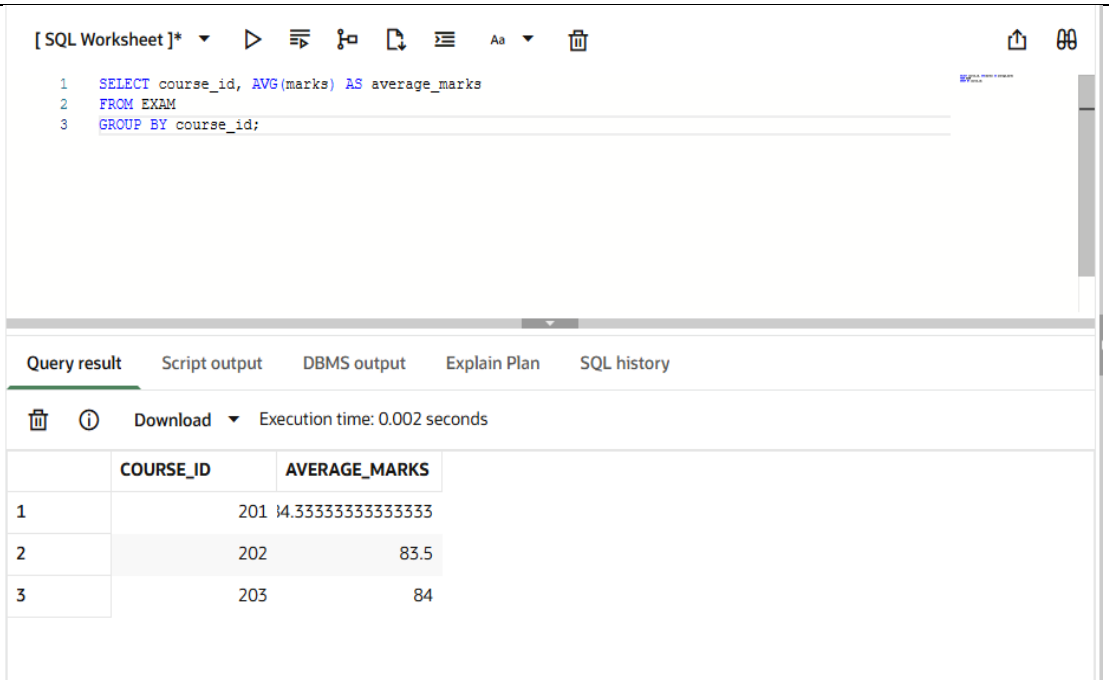

| | COURSE_ID | TOP_MARKS |
|---|---|---|
| 1 | 201 | 90 |
| 2 | 202 | 92 |
| 3 | 203 | 88 |

- **Count: -**

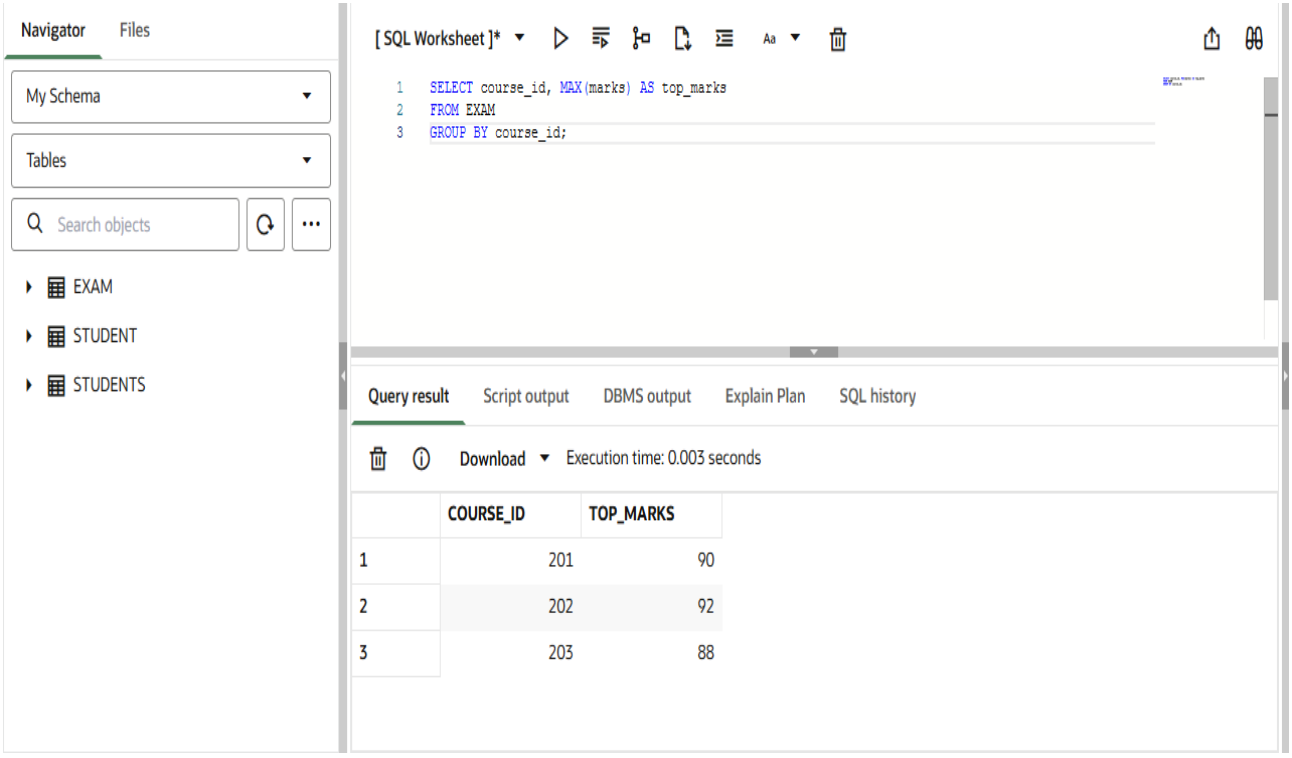



| | COURSE_ID | STUDENTS_APPEA |
|---|---|---|
| 1 | 201 | 3 |
| 2 | 202 | 2 |
| 3 | 203 | 2 |

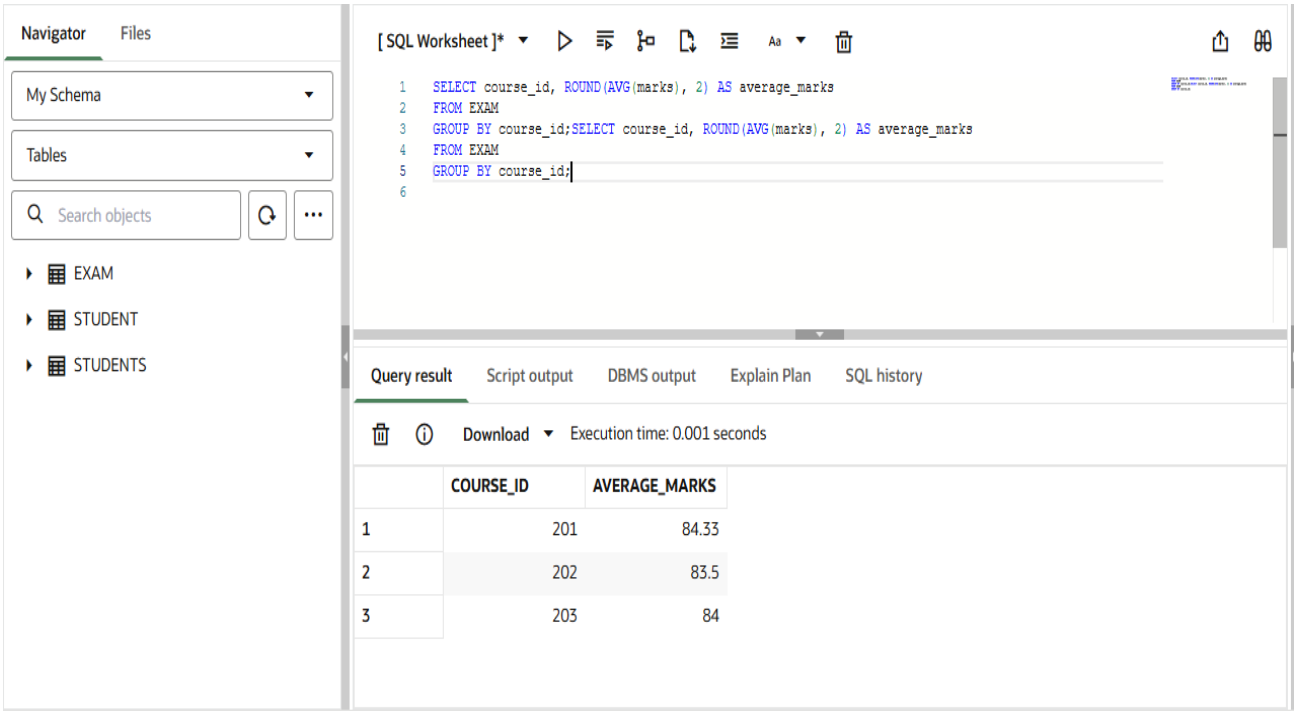
- **Average: -**



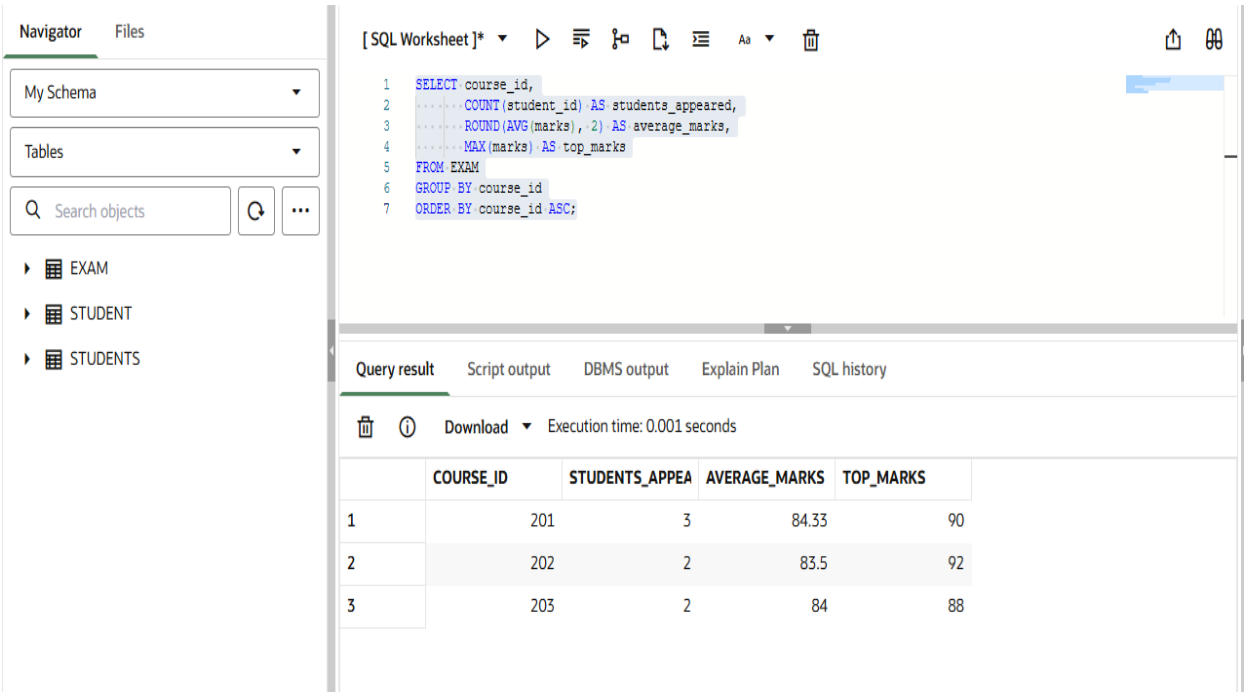

- **Maximum: -**



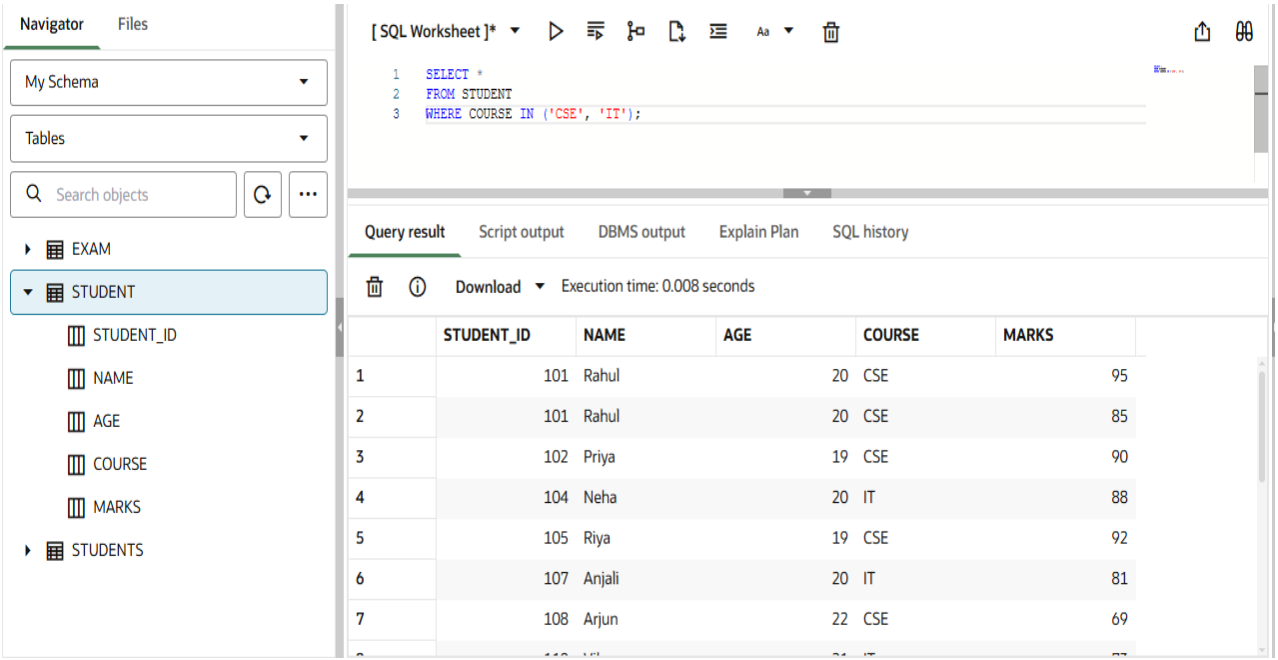
- Round: -



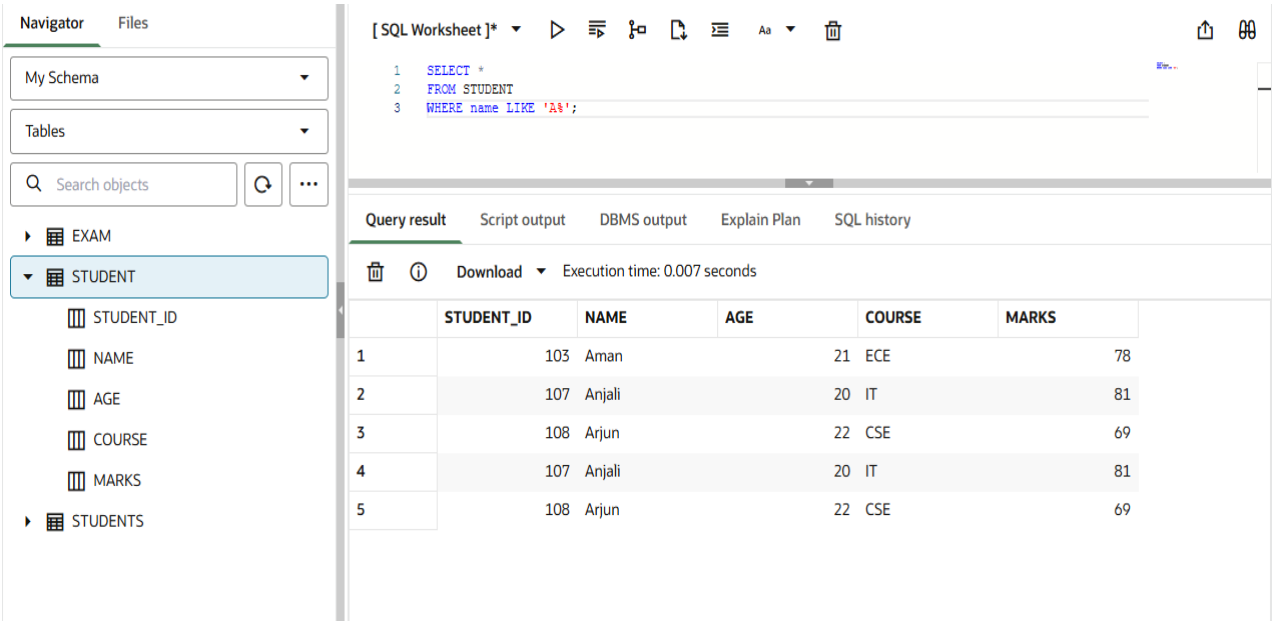

- Final Table: -


Navigator
Files
My Schema
Tables
Search objects
EXAM
STUDENT
STUDENTS
[ SQL Worksheet ]*
SELECT course_id,
COUNT(student_id) AS students_appeared,
ROUND(AVG(marks), 2) AS average_marks,
MAX(marks) AS top_marks
FROM EXAM
GROUP BY course_id
ORDER BY course_id ASC;
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.001 seconds
COURSE_ID
STUDENTS_APPEA
AVERAGE_MARKS
TOP_MARKS
1 201 3 84.33 90
2 202 2 83.5 92
3 203 2 84 88

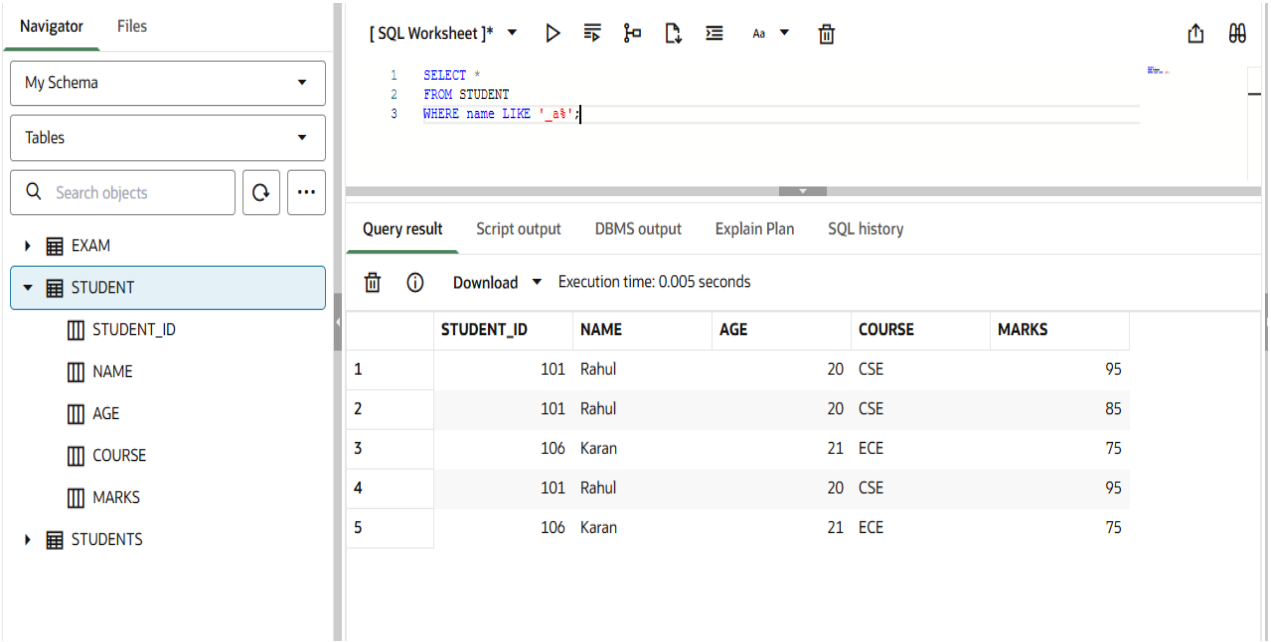
- IN: -


Navigator
Files
My Schema
Tables
Search objects
EXAM
STUDENT
STUDENT_ID
NAME
AGE
COURSE
MARKS
STUDENTS
[ SQL Worksheet ]*
SELECT *
FROM STUDENT
WHERE COURSE IN ('CSE', 'IT');
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.008 seconds
STUDENT_ID
NAME
AGE
COURSE
MARKS
1 101 Rahul 20 CSE 95
2 101 Rahul 20 CSE 85
3 102 Priya 19 CSE 90
4 104 Neha 20 IT 88
5 105 Riya 19 CSE 92
6 107 Anjali 20 IT 81
7 108 Arjun 22 CSE 69

- Like: -


Navigator
Files
My Schema
Tables
Search objects
EXAM
STUDENT
STUDENT_ID
NAME
AGE
COURSE
MARKS
STUDENTS
[ SQL Worksheet ]*
1 SELECT *
2 FROM STUDENT
3 WHERE name LIKE 'A%';
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.007 seconds
STUDENT_ID NAME AGE COURSE MARKS
1 103 Aman 21 ECE 78
2 107 Anjali 20 IT 81
3 108 Arjun 22 CSE 69
4 107 Anjali 20 IT 81
5 108 Arjun 22 CSE 69

- Like(_a): -


Navigator
Files
My Schema
Tables
Search objects
EXAM
STUDENT
STUDENT_ID
NAME
AGE
COURSE
MARKS
STUDENTS
[ SQL Worksheet ]*
1 SELECT *
2 FROM STUDENT
3 WHERE name LIKE '_a%';
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.005 seconds
STUDENT_ID NAME AGE COURSE MARKS
1 101 Rahul 20 CSE 95
2 101 Rahul 20 CSE 85
3 106 Karan 21 ECE 75
4 101 Rahul 20 CSE 95
5 106 Karan 21 ECE 75